Research Document on NOVA Buildathon 2026

Contents

Attention Lapse Detection

COG-BCI Dataset

The **COG-BCI database** [1] is a multi-session, multi-task EEG dataset designed for research on **passive brain-computer interfaces** (pBCI). Each subject performed the same battery of cognitive tasks across three sessions while **63 EEG channels** were recorded at **500 Hz** with a **common-average reference**. For every condition, the dataset also provides trial-level **behavioral logs**, which makes it possible to align neural activity with overt performance.

Dataset Structure

The local copy follows a **BIDS-like layout**, organized per subject (`sub-XX`) and session (`ses-S1`–`ses-S3`):

- `eeg/` – raw EEGLAB files (`.set` header + `.fdt` binary data) for each condition;
- `behavioral/` – trial-level behavioral logs (`.mat`) mirroring the same conditions;
- `chanlocs/` – channel names and 3D coordinates.

Each session contains the same task battery: the N-back task (0/1/2-back), the Flanker task, the Multi-Attribute Task Battery (MATB), the Psychomotor Vigilance Task (PVT), and resting-state blocks (eyes open / eyes closed) at the start and end of the session.

Structure of Data on Psychomotor Vigilance Task

The **Psychomotor Vigilance Task** (PVT) is a classic **sustained-attention** paradigm. Participants watch a screen and press a button as soon as a stimulus appears. Stimuli are presented at **pseudo-random inter-stimulus intervals** of 2–10 s, so the task cannot be solved by rhythm or anticipation; performance therefore tracks the current level of vigilance.

The primary outcome is the **reaction time** (RT) on each trial. Slow responses and **lapses** are the hallmark of reduced vigilance, which is observed under sleepiness, fatigue, and **mind wandering**. When the mind wanders, attention is diverted away from the stimulus, so the participant reacts late or not at all – precisely the lapse signature that the PVT measures. PVT performance is therefore a standard **behavioral proxy** for mind-wandering episodes, and, combined with the simultaneously recorded EEG, it allows searching for the neural correlates of attentional drift.

The PVT EEG is stored in **EEGLAB format**: a `.set` header file together with a `.fdt` binary file that holds the raw samples. Inspecting the MATLAB struct of `PVT.set` gives the following picture:

```
EEG =
  filename: 'PVT.set'   datfile: 'PVT.fdt'
  nbchan:  63           trials: 1
  pnts:    306060       srate:  500
  xmin:    0            xmax:   612.118
  data:    [63×306060 single]
  ref:     'common'
  event:   [272×1 struct]
```

The various fields have the following meaning:

- `nbchan: 63`, `srate: 500` – the recording contains **63 EEG channels** sampled at **500 Hz**.
- `trials: 1`, `pnts: 306060`, `xmax: 612.118` – a **single continuous recording** of 306060 sample points, i.e. roughly **612 seconds (~10.2 minutes)**, that is **not** segmented into epochs.
- `data: [63×306060 single]` – the raw amplitude values (in microvolts, single-precision floats) laid out as **channels** × **time points**.

- `ref: 'common'` – the signal has already been **average-referenced**.
- `chanlocs: [1×63 struct]` – the channel locations, matching `chanlocs/get_chanlocs.txt`.
- `event: [272×1 struct]` – event markers for the 90 PVT trials (stimulus onsets and responses) together with session start/end markers; these events make it possible to **segment the continuous signal into trials**.
- `datfile: 'PVT.fdt'` – the companion binary file containing the actual samples.

Corrupted Behavioral Log

The behavioral log `PVT.mat` describes the same 90 trials at the performance level. In theory, taken together, `PVT.set` / `PVT.fdt` provide the continuous neural signal while `PVT.mat` provides the per-trial behavioral ground truth. However, an inspection on the `PVT.mat` files using **MD5** shows that the files seem to be **identical across sessions and subjects**, which is unexpected.

A further inspection of the EEG files shows that `sub-17` and `sub-27`, `sub-25` and `sub-28` have identical EEG recordings for the PVT. Those four subjects are removed from the analysis, leaving 25 subjects with valid data.

Since we are training for a model that detects lapses from EEG, we need to have the reaction times for each trial in order to label the trials as **lapse** or **non-lapse**. Hence, we need to **reconstruct the behavioral log** from the EEG event markers.

We use the `mne` package to read the EEG files as `raw` and extract the event markers via `raw.annotations`. The event labeled as `13` indicates the **stimulus onset** and the event labeled as `14` indicates the **response**. The difference between the two timestamps gives the **reaction time** for each trial.

Training EEGNet Using the COG-BCI Dataset

Data Labeling

In order to obtain a clean training set, each PVT trial must be labeled and its EEG segment extracted before being fed to the model.

We use the `mne` package to read the raw EEG recording as a `Raw` object and extract the event markers through `raw.annotations`. As described above, the annotation labeled `13` marks the **stimulus onset** and the annotation labeled `14` marks the **response**; the difference between the two timestamps gives the **reaction time** of the trial.

A trial is labeled as a **lapse** when the participant fails to respond in time. A global threshold such as the classic 500 ms does not generalize here: an analysis of the data shows that for some participants all reaction times are below 500 ms. We therefore use a **per-participant, data-driven threshold**: for each participant, the trials whose reaction times fall in the **longest 10%** are labeled as lapses, and all the others as non-lapses.

Since we want the model to learn the neural pattern preceding an attention lapse, the EEG of interest is the one **right before the stimulus**. For every trial we extract the **2000 ms of signal ending 100 ms before the stimulus onset**; the 100 ms gap ensures the segment does not include the stimulus-evoked response itself.

Because the extracted segment sits immediately before the stimulus, it must not be contaminated by the **remnant of the previous response or error trial**. A trial is therefore kept only if the previous response/error happened at least **2000 ms + 200 ms** before the current stimulus onset: 2000 ms so that the whole segment is clear of the previous event, plus a **200 ms buffer** so that any residual activity from the last response/error has time to disappear. Trials that do not satisfy this spacing requirement are discarded.

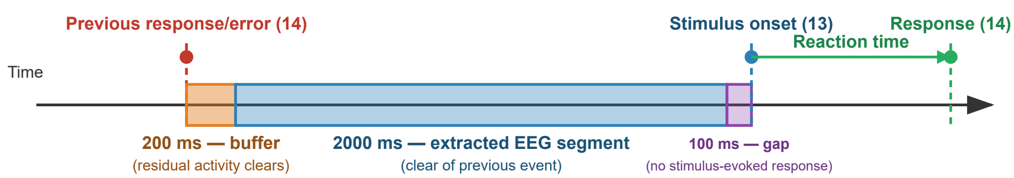

Figure 1: Spacing criterion for keeping a trial: the previous response/error must occur at least **2000 ms + 200 ms** before the current stimulus onset, so that the extracted 2000 ms EEG segment is not contaminated by residual activity.

Model Input

Before being fed to the model, the extracted segments are prepared as input. Depending on the session, the recordings contain **63 or 64 channels**, one of which is an **ECG channel**; we therefore remove that channel, which reduces the data to **62 channels** (63 to 62, 64 to 63). To keep a single consistent input shape across all subjects and sessions, the network is fed exactly these **62 channels**.

The raw signal, originally recorded at **500 Hz**, is **downsampled to 128 Hz**. For every trial the model receives **2 s of signal** (the 2000 ms segment ending 100 ms before the stimulus onset), which corresponds to **256 time samples** at 128 Hz. The final input to the network is therefore a **62-channel × 256-sample** window.

Training EEGNet

The **EEGNet** architecture [2] is a compact convolutional neural network designed for EEG-based brain-computer interfaces. It learns temporal features with regular convolutions and spatial features with **depthwise and separable convolutions**, which makes it well suited to small datasets with limited training data. We **replicated the model in PyTorch**.

To obtain an unbiased estimate of cross-subject generalization, we use **leave-one-subject-out** (LOSO) cross-validation. There are **25 available subjects** (4 out of the original 29 were removed because their EEG recordings were identical); each fold leaves one subject out, trains the model on the remaining **24 subjects**, and evaluates on the held-out subject.

Because lapses are the minority class (only the **longest 10%** of reaction times are labeled as lapses), the training set is imbalanced between the positive (lapse) and negative (non-lapse) trials. To deal with this potential imbalance, we train with **focal loss**, which down-weights the well-classified majority trials so that the model focuses on the harder, minority lapse trials.

Since the classes are imbalanced, accuracy would be a misleading metric. We therefore use the **F1 score** to evaluate the model's performance.

Result

- **Multiple Windows**:
  To increase the amount of training data, we first collected **multiple windows per trial**: **4 windows of 2 s each**, ending at **100 ms, 200 ms, 300 ms and 400 ms before the stimulus** in attempt to increase the number of training samples. This augmentation **did not help** the model – it actually **decreased** the F1 score: as we reduced the number of windows from 4 to 2 to 1, the F1 score increased.

Windows	F1 score
4	0.578 ± 0.059
2	0.584 ± 0.061
1	0.591 ± 0.059

- **Focal Loss**:
  We limited the **ratio of positive and negative samples** in the training set and adjusted the focal loss accordingly. **Changing the sample ratio produced no improvement**: for a single window, the F1 score stays around **0.59**.

References

[1] M. F. Hinss, E. S. Jahanpour, B. Somon, L. Pluchon, F. Dehais, and R. N. Roy, *COG-BCI database: A multi-session and multi-task EEG cognitive dataset for passive brain-computer interfaces.* (Jul. 2022). Zenodo. doi: 10.5281/zenodo.6874129.

[2] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, p. 56013, Jul. 2018, doi: 10.1088/1741-2552/aace8c.